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Lysergide Treatment of Schizophrenic and Nonschizophrenic Alcoholics

A Controlled Evaluation[1]

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SUMMARY. *Of 75 alcoholics treated with lysergide, 23 were diagnosed as schizophrenics. The nonschizophrenics had better reactions to the drug and more of them were abstinent at a 1-year follow-up than the schizophrenics and than alcoholics treated by other methods.*

A SUBSTANTIAL NUMBER of studies evaluating the effectiveness of lysergide (LSD) in the treatment of alcoholism have appeared in the literature since 1953 when Hoffer and Osmond first used it in alcoholics at the University Hospital at Saskatoon, Saskatchewan, with the intention of giving them an experience that would simulate delirium tremens. At that time it was hoped that such a safely controlled experience would motivate the patient positively by giving him a preview of what might happen in the future. This early idea was soon abandoned and emphasis was shifted toward the psychedelic aspects of the drug (1). Smith, who joined Hoffer's staff at about this time, carried on the work that had been started there, and in 1958 published his first

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report, based on 29 patients, and concluded that lysergide had beneficial effects (2).

In the next 7 years research with lysergide in alcoholism was carried on almost exclusively in the Canadian Provinces and resulted in a series of reports all of which concluded that it was either beneficial or at least very promising (2–9).[4] Some of these studies (3, 6, 9) have been sharply criticized by Smart and Storm (10) for their lack of adherence to minimal standards for evaluating new drugs. These standards should include random assignment of patients to control or treatment groups, blind administration of treatment, blind and objective assessment of outcome at fixed intervals after treatment, control for placebo effects, and control by comparison with some other treatment.

Jensen (4) and Jensen and Ramsay (5) reported the first projects that used controls in 1962 and 1963, respectively, and found statistically significant gains for the lysergide-treated group. The control group was described as patients admitted to the hospital when no beds were available to the study group and received individual treatment by other psychiatrists. The use of controls in this report was a good step forward but a reasonable doubt still existed as to whether the control group had a comparable treatment.

In 1966, Smart and his co-workers (11) published what is probably the best methodological model of lysergide research in alcoholism to appear up to that time. They reported on 30 patients who were divided into 3 groups: 10 received lysergide, 10 controls received ephedrine sulphate, and 10 received a comparable treatment without drugs. The historical significance of this study was the attempt to deal with the problem of controls in two ways: a control by comparable experience minus lysergide and a control for possible placebo effect. Other desirable features in this study were a specific follow-up period of 6 months, plus or minus about 2 weeks, and a method of measuring improvement by assessing gain in abstinence for each individual based on pre- and posttreatment measures. No differences were found between the treated and control groups; however, the number of cases was small and significant differences could have been obscured. Whatever the limitations of the study, it showed that the effects of lysergide are not so dra-

[4] Also, ABLES, M. F. and ENG, E. W. Group treatment of chronic alcoholism with LSD. Presented at the 12th Annual Conference of V.A. Cooperative Studies in Psychiatry, Denver, 1967.

matic or easily predictable that they can be demonstrated with a few cases and no controls.

A small pilot study by Van Dusen et al. (12), and a later continuation with more cases (13), also yielded negative results. This study was somewhat different from most of the others in that the patients treated with lysergide and the controls were women, and also in that it attempted to assess the effect of multiple experiences rather than the "single overwhelming experience." The use of multiple experiences was not, in this case, an attempt to increase the frequency of psychedelic reactions but was a study of this variable by random assignment of patients to one, two or three lysergide sessions.

A recent study, described in a progress report by Ables and Eng,[4] was based on 63 lysergide-treated patients and 59 controls at the Veterans Administration Hospital in Lexington, Ky. Preliminary results indicated that during a 10 to 15 month postdischarge period there was significantly more improvement in drinking control and general adjustment in the lysergide-treated group than in the controls.

From the vantage point in time it can now be said that the early researchers, who collectively treated a rather large population of patients but did not utilize adequate controls, did so on the basis of a subjective impression of what could be expected in treating alcoholics. Their enthusiasm was easily generated by seeing many patients profoundly moved by the experience and going back to society with new attitudes, insight, hope and enthusiasm. The subsequent negative findings in some of the controlled studies, however, seem to indicate that the striking changes may only be a transient state that eventually fades when the alcoholic patient returns to society and old patterns regain their power.

Our study began in September 1965, and data collection ended in April 1968, when 1-year follow-up information was available on the last lysergide-treated case. During this period we became increasingly aware of the methodological difficulties of evaluating lysergide and often wished we could have changed some aspects of the experimental design to which we were committed. On the positive side, we continued to feel that we were in a good position to attempt an evaluation of lysergide because we had valid follow-up information on a large number of patients who had gone through

our regular 90-day rehabilitation program and they provided us with a good comparison group.

METHOD

Dosage and Physical Setting

Our method is one that has been described in other studies as a "single overwhelming experience." At approximately 8:30 AM the patient received 500 μg of lysergide in a specially furnished room. Previous research had established the value of a proper setting for the experience. The room used in this study incorporated most of the suggestions that appeared in the literature: It was approximately 15 by 20 feet in size with colorful drapes and pictures. A stereo record player with a wide selection of music was available upon the patient's request. Additional props included an album of colored pictures, artificial flowers, and a hand mirror. This particular room was selected because it was relatively free from the usual ward noises. If the patient wanted to look out of the window, he had a pleasant view of the wooded hospital grounds and mountains in the distance.

Selection of Subjects and Controls

The lysergide subjects were 75 volunteers selected from the population of patients in the 90-day Alcoholic Rehabilitation Program. Soon after the patient was admitted to the Program he was told that a research project evaluating possible therapeutic benefits of lysergide was being conducted and he could volunteer to take the drug. If the patient volunteered he was paired with another volunteer who entered the Program at about the same time. The participants were told that when they were approximately halfway through the Program (about 45 days) one of them would be selected for the lysergide treatment by the toss of a coin. All of the volunteers were asked to begin writing an autobiography according to a prescribed format and to include in this written material their reasons for taking the drug, personal problems, and what they expected to receive from the experience. The autobiographical material had to be completed before the coin was tossed.

A second group of controls was a large population of patients who had passed through the Program and were part of an ongoing follow-up evaluation. This large group, which provided a stable measure of what was being achieved by the regular Program, was similar in composition to the lysergide controls but different in one respect: They either had passed through the Program before the lysergide project was started or they did not volunteer for the lysergide treatment. The controls who volunteered to take lysergide but did not receive it, or received a placebo substitute, will be referred to as control group I, and the second group (regular Program participants) will be called control group II.

Composition of Lysergide and Control Groups

No attempt was made to screen out or control the number of schizophrenic alcoholics who volunteered for lysergide therapy. There were 23 schizophrenics in the group that received lysergide, 30 in control group I, and 60 in control group II. Patients were diagnosed as schizophrenic on the bases of past medical history, MMPI profile and clinical appearance. None of these patients were acutely psychotic at the time of the study or they would not have been admitted to the Alcoholic Rehabilitation Program. We were able to apply the diagnosis of schizophrenia with a high level of certainty in about 65% of these patients; the remainder were more ambiguous and other diagnosticians might have labeled them as schizoid personalities.

Table 1 shows the psychiatric subclassification of the nonschizophrenic patients at the beginning of the study. All of the groups appear to be reasonably equated by diagnostic subtype.

Table 2 presents the composition of the nonschizophrenic groups at the beginning and end of the study according to a number of variables. The data appear to support our assumption that loss of cases during the follow-up year did not selectively distort composition of the groups on these variables. A similar accounting of the schizophrenic alcoholics was not made because the number was too small for meaningful analysis.

Conducting the Lysergide Experience

Patients who volunteered to serve as subjects were given an explanatory lecture on the nature of lysergide and were asked to read a selection of suitable writings. This preparatory step was aimed at reducing any strong apprehensions or correcting any gross misunderstandings on the part of patients.

During the first 7 hours of the experience a nursing assistant or nurse selected by the patient stayed with him continuously and made written notes. This trained participant functioned in a supportive role if necessary but did not probe, interpret or try to direct the experience. The psychiatrist who administered the lysergide visited the patient several times during the experience and interacted with him in the same manner.

TABLE 1.—*Psychiatric Subclassification of Nonschizophrenic Lysergide-Treated and Control Subjects*

	Lysergide		*Control I*		*Control II*	
	N	*%*	*N*	*%*	*N*	*%*
Neurotic	34	65	28	62	83	67
Antisocial personality	8	15	9	20	12	10
Uncomplicated alcoholism	5	10	3	7	8	7
Chronic brain syndrome	0	0	1	2	4	3
Drug addiction	1	2	1	2	5	4
Borderline mental deficiency	0	0	0	0	5	4
Not evaluated	4	8	3	7	6	5
Totals	*52*	*100*	*45*	*100*	*123*	*100*

TABLE 2.—*Composition of the Nonschizophrenic Alcoholic Lysergide and Control Groups at the Beginning (3 Months) and End of the Study (1 Year)*

	Lysergide		*Control I*		*Control II*	
	3 mo. (*N*=48)	1 yr (*N*=32)	3 mo. (*N*=33)	1 yr (*N*=26)	3 mo. (*N*=101)	1 yr (*N*=78)
Mean age	43	45	43	45	45	45
Otis IQ mean	101	102	100	98	93	92
Mean years of problem drinking	12.8	11.8	11.4	12.1	10.2	10.7
Mean days in program	87	88	87	84	72	74
Mean months self-enforced sobriety	14	16.1	10.4	11.1	10.3	11.3
Type of Discharge (%)						
Disciplinary	4	3	6	4	6	4
Against medical advice	10	9	3	8	38	39
Completed program	86	88	91	88	56	57
Marital Status (%)						
Single	4	6	3	4	11	13
Married	44	49	52	50	56	55
Separated	13	12	15	19	3	4
Divorced	35	28	27	27	28	28
Widowed	4	5	3	0	2	0
Previous Treatment (%)						
None	10	16	6	8	22	24
Physician	46	31	52	50	41	37
Hospital without program	60	63	70	62	51	41
Hospital with program	35	28	30	15	24	26
A.A. participation	65	63	67	65	48	46

The patient was not encouraged to talk extensively about his experience at that time, but the next day he spent an hour in his regular therapy group giving a narrative account and analyzing the meaning of his experience.

No additional drugs were used to terminate the experience or to ensure a good night's sleep afterward. If the patient chose to stay up all night after the experience and talk with interested patients or personnel, he was allowed to do so.

Assessing the Experience

Individual differences in the lysergide experience were assessed by a psychometric method and a subjective classification. The psychometric method utilized the Blewett and Chwelos Scales for the Assessment of Psychedelic Reactions.[5] Scale 1 was completed by the patient the day

[5] BLEWETT, D. B. and CHWELOS, N. Handbook for the therapeutic use of lysergic acid diethylamide-25: individual and group procedures. [Unpublished.]

after the session. It is a self-inventory consisting of 66 items that allow the patient to indicate various perceptual, emotional, conceptual and transcendental aspects of his experience. The degree to which the patient achieved a good therapeutic response in the conventional sense of the word was assessed from 22 items of this scale. Patients scored high if they said they were not unduly frightened, felt the experience was very important, felt a trust in other people rather than a continuous suspicion, etc. In the extremely good reactions they achieved a transcendental feeling of unity with the universe. Scale 2 was administered after a passage of time that varied from 30 to 45 days. It inquired about the continuation of new attitudes about the self and others. A person who scored very high would be regarded as having a very marked, desirable personality change that was still present 30 to 45 days after the experience.

The subjective classification of the experience was collectively based on (*1*) observations of the patient during the session, (*2*) his narrative account the next day, (*3*) his written account, and (*4*) observation of any changes in the patient during the remainder of his stay in the Program. "Intellectual" and "emotional" insight were judged as good, moderate or poor, depending on the degree to which the patient achieved a correlation of past experiences with present psychopathology and demonstrated some degree of correction of faulty emotional reactions by actual behavior in his posttreatment interpersonal relations. Those patients who clearly showed a high degree of these intellectual, emotional and behavioral changes were judged to have had a "good" experience. At the other extreme, those who showed a marked poverty of these elements were rated as "poor." The mixed or poorly defined cases were rated as having a "moderate" experience. A judgment was also made regarding the "spiritual" quality and content of the experience. To those patients who experienced it most clearly, it was a feeling of kinship with all forms of life accompanied by an emotional experience of a higher power and a paradoxical feeling of being all-important and yet insignificant.

Postdischarge Evaluation of Drinking Control

Shortly before the patient was discharged from the Program he received an interview in which the contents of a follow-up questionnaire were reviewed with him. At this time permission was obtained for the investigators to contact by telephone his spouse or some other individual who could validate the information he provided. An effort was made to enlist the patient's cooperation and remove any fears that he must send in a "good" report or disappoint the staff and perhaps lose some of his veteran's benefits. The most important part of the questionnaire is the patient's self-rating on a Drinking Adjustment Scale, made by checking categories ranging from complete abstinence to drinking heavily enough to require medical care. The information from the patient was obtained in six categories of drinking control. These were later collapsed into

four categories (Chart 1) that provided less discrimination but were probably more valid.

The patient received this questionnaire 3 months, 6 months and 12 months after discharge from the hospital. Those who did not return their questionnaires promptly were contacted by telephone. We tried to contact the validation source by telephone in all cases when drinking was described as "total abstinence" or "much improved" and also in some of the cases in Category *C* (questionable improvement). It obviously was not necessary to validate reports in Category D (unchanged). In the 12-month follow-up population we were able to validate 50% of the lysergide group, 37% of control group I and 54% of control group II. Our validation method was developed and practiced in accumulating the data for control group II. Several years of experience showed us that there were almost no cases of obvious falsification and we therefore have a very high level of confidence in the nonvalidated data. It may seem strange that such a high level of veracity would be found in a group of alcoholics, but it should be remembered that our predischarge interview prepared them for honest participation and the patient knew we might try to corroborate his statements through the validation source.

Assessing Lysergide Recrudescence

A recrudescence of the lysergide state at some time after the experience has been noted as an infrequent but well-established characteristic of the drug. Like the lysergide state itself, recrudescence is unpredictable and varied in content. Previous studies of lysergide therapy of alcoholism did not assess this effect or did not report it. We did not originally plan to study recrudescence; however, after a few patients reported brief recurrences of the lysergide state when they returned their follow-up questionnaires, we began to make a formal inquiry by requesting this additional information. The patients were told in the follow-up form that recurrence of the lysergide state happens occasionally and that we wanted his report on the frequency and nature of any recurrences.

RESULTS

Classification of the LSD Experience

Table 3 shows the score distributions on the Blewett and Chwelos scales. The nonschizophrenic group achieved a higher median score on both scales, indicating a slightly more therapeutic lysergide ex-

CHART 1.—*Drinking Adjustment Scale*

A. Complete abstinence.

B. Much improved—only one drink or drinking episode during the rating period.

C. Slight or questionable improvement—drinking occasionally or regularly but having no employment, social or marriage problems.

D. Unchanged—drinking heavily and as a result having employment, social, marriage or medical problems.

TABLE 3.—*Distribution of Blewett and Chwelos Scale Scores of Nonschizophrenic and Schizophrenic Alcoholics*

Scale 1	*Nonschizophrenic*		*Schizophrenic*	
Score	*N*	*%*	*N*	*%*
21–22[a]	7	13	1	4
19–20	12	23	3	13
17–18	12	23	3	13
15–16	11	21	6	26
13–14	4	8	6	26
11–12	3	6	1	4
9–10	3	6	1	4
7–8	0	0	2	10
Totals	*52*	*100*	*23*	*100*

Scale 2	*Nonschizophrenic*		*Schizophrenic*	
Score	*N*	*%*	*N*	*%*
30–33	17	35	4	18
26–29	6	13	4	18
22–25	6	13	4	18
18–21	4	9	3	14
14–17	4	9	4	18
10–13	6	13	1	5
6–9	2	4	0	0
7–8	2	4	2	9
Totals	*47*	*100*	*22*	*100*

[a] Higher scores are most and lower scores least therapeutic. Median scores: nonschizophrenic groups, Scale 1, 17.5, Scale 2, 24.5; schizophrenic groups, Scale 1, 15.0, Scale 2, 20.5.

perience and a stronger tendency to regard the experience positively with the passage of time. A chi-square test based on the frequency of Scale 1 scores above and below a reference score of 16.5 was found to be statistically significant beyond the .05 level of confidence. A similar test of Scale 2 was not significant. Table 4 shows that the nonschizophrenic patients had a slightly higher percentage of "good" experiences and a slightly lower percentage of "poor" experiences. The distribution for spiritual content was almost identical. Our expectation that the nonschizophrenic patients would have a better experience was not strongly borne out, although the differences are in the predicted direction. Two of our lysergide-treated patients gave us concern because they lost contact with reality during the height of the experience and acted bizarrely. Later they could not completely recall what had happened. Both of these patients had nonschizophrenic diagnoses. Some of the patients reported recrudescence effects shortly after the experience and also

TABLE 4.—*Classification of the Lysergide Experience*

	Reaction						Spiritual Content			
	Good		*Moderate*		*Poor*		*Yes*		*No*	
	N	%	*N*	%	*N*	%	*N*	%	*N*	%
Schizophrenic alcoholics	8	35	8	35	7	30	8	35	15	65
Nonschizophrenic alcoholics	20	40	19	39	11	22	17	34	33	66

during the follow-up part of the study. There were no known instances of serious detrimental effects.

Effects of Lysergide upon Postdischarge Drinking Control

When this investigation was started we did not intend to make a specific comparative evaluation of schizophrenic and nonschizophrenic alcoholics, but to have approximately equal proportions of these diagnostic types in the lysergide-treated and control groups. Preliminary inspection of the data, however, indicated a marked negative effect of lysergide upon schizophrenic alcoholics (Table 6) and the study had to be retitled even though we had a relatively low number of schizophrenics in our groups. Our data consist of postdischarge drinking ratings on six different groups of alcoholics: (*1*) nonschizophrenic lysergide-treated group, (2) nonschizophrenic control group I, (3) nonschizophrenic control group II, (*4*) schizophrenic lysergide-treated group, (5) schizophrenic control group I, and (*6*) schizophrenic control group II. Tables 5 and 6 give numbers and percentages of patients in four categories of drinking control at three different periods following discharge. Table 7 is a summary of the chi-square comparisons of all the groups based on a twofold complete-abstinence versus nonabstinence classification rather than the four-part classification in Tables 5 and 6. Inspection of the distributions of drinking control in these tables indicate that when lysergide benefits occur they tend to effect complete abstinence rather than a reduced consumption of alcohol such as would be reflected in our "much improved" or "slight questionable improvement" categories. A two-part classification, therefore, amplified the differences and our chi-square comparisons were made on this basis. In Table 7 only those chi squares that are statistically significant or appear to approach significance are given. Cells with a chi square of less than 1.0 are left blank. All the groups

TABLE 5.—*Postdischarge Drinking Control of Lysergide-Treated Patients and Control Groups of Nonschizophrenic Alcoholics*

Follow-Up Period	*Reported Cases (N)*	Drinking Adjustment Classification A*		B*		C*		D*		*Data Loss*†		*Total N*
		N	*%*	*N*	*%*	*N*	*%*	*N*	*%*			
Three months												
Lysergide	48	24	50	6	12	9	19	9	19	4	8	52
Control I	33	10	30	7	21	7	21	9	28	12	27	45
Control II	101	40	39	16	16	30	30	15	15	22	18	123
Six months												
Lysergide	42	15	36	5	12	11	26	11	26	10	19	52
Control I	29	5	17	6	21	8	28	10	34	16	36	45
Control II	87	32	37	7	8	27	31	21	24	36	29	123
Twelve months												
Lysergide	32	14	44	4	13	9	28	5	15	20	38	52
Control I	26	3	11	5	19	9	35	9	35	19	42	45
Control II	78	24	31	8	10	34	43	12	16	45	37	123

* A = complete abstinence; B = much improved; C = slight or questionable improvement; D = unchanged. See Chart 1.
† Cumulative data lost because patients could not be located, died, or were removed from the study for methodological reasons (see text).

TABLE 6.—*Postdischarge Drinking Control of Lysergide-Treated Patients and Control Groups of Schizophrenic Alcoholics*

Follow-Up Period	*Reported Cases (N)*	DRINKING ADJUSTMENT CLASSIFICATION A*		B*		C*		D*		*Data Loss†*		*Total N*
		N	*%*	*N*	*%*	*N*	*%*	*N*	*%*			
Three months												
Lysergide	16	3	19	0		8	50	5	31	7	30	23
Control I	23	8	35	5	22	6	26	4	17	7	23	30
Control II	43	12	28	14	33	10	23	7	16	17	28	60
Six months												
Lysergide	15	3	20	3	20	4	27	5	33	8	35	23
Control I	21	7	33	2	9	6	29	6	29	9	30	30
Control II	37	13	35	7	19	11	30	6	16	23	38	60
Twelve months												
Lysergide	12	2	17	0		2	17	8	66	11	48	23
Control I	17	4	24	0		5	29	8	47	13	53	30
Control II	27	7	27	2	7	8	29	10	37	33	55	60

* A = complete abstinence; B = much improved; C = slight or questionable improvement; D = unchanged. See Chart 1.
† Cumulative data lost because patients could not be located, died, or were removed from the study for methodological reasons (see text).

TABLE 7.—*Chi-Square Comparison of All Alcoholic Groups Based on Abstinence–Nonabstinence Classification of Drinking Control at 1 Year*[a]

	Nonschizophrenic			*Schizophrenic*		
	1	*2*	*3*	*4*	*5*	*6*
Group 1, nonschizophrenic, lysergide treated ($N=32$)	–	7.19*	2.25	2.78	1.96	2.03
Group 2, nonschizophrenic, control group I ($N=26$)		–				
Group 3, nonschizophrenic, control group II ($N=78$)		3.75	–			
Group 4, schizophrenic, lysergide treated ($N=12$)				–		
Group 5, schizophrenic, control group I ($N=17$)					–	
Group 6, schizophrenic, control group II ($N=27$)		1.78				–

[a] Only significant chi squares and those approaching it are shown. Chi square for .05 level of confidence with df 1 = 3.84. The superior group is indicated by the group number on the left.
* Significant at .01 level of confidence.

in Tables 5 and 6 show a typical erosion of drinking control in the course of a year. Whatever benefits occur from lysergide or the regular Alcoholic Rehabilitation Program fade in the course of a year and this is reflected in the decreasing percentage of individuals who were abstaining and a parallel increased percentage in the other categories. The greatest gain is shown by the nonschizophrenic lysergide-treated patients. They have the highest percentages of patients abstaining at the end of 3 months and maintain this superiority throughout the year. Table 7 shows that the nonschizophrenic lysergide-treated alcoholics were superior to all the other groups in varying degrees. The difference between them and the nonschizophrenic control group I was significant beyond the .01 level of confidence. This statistically significant difference, however, provides only a questionable basis for assuming lysergide benefits, since the nonschizophrenic control group I was not representative of what was being achieved by the regular Alcoholic Rehabilitation Program (control group II). We were very surprised to find that nonschizophrenic control group I showed the poorest drinking control of all the groups at the end of a year. The statistically significant difference between the nonschizophrenic lysergide-treated patients and control group I suggests that we may have inadvertently introduced a strong deprivation effect in which the nonschizophrenic patients of control group I felt disappointed because they did not receive the "magical drug." Although control group I appears to be of no value in evaluating lysergide benefits, the data

illustrate the complexities and difficulties of achieving control over the placebo effect of a drug that has spectacular mind-altering properties, and where research is contaminated by expectations of benefit.

There was an unfortunate loss of cases in all the groups during the 1-year period, as shown on the right side of Tables 5 and 6. Most of the cases were lost because we could not keep in contact with the patients despite our best efforts. A few patients died and about one-third had to be removed for methodological reasons. Since we were trying to evaluate the effects of lysergide added to the regular treatment program, we could not retain a patient in the study who was readmitted to our hospital or some other hospital for additional treatment of the kind he had received in the Program. We also had to assume that a patient was free to drink. If we learned that a patient was incarcerated for a long time or hospitalized, he was dropped. Because of the variety of situations that can arise, we could not apply a precise set of rules for dropping a case from the study but had to make a separate judgment each time. We assumed that removing these cases from all the groups enabled us to assess the effects of the regular program and the lysergide treatment more accurately by reducing the confounding effects of additional variables. We assumed that the loss of data did not introduce a fatal defect into the study because the percentage of cases lost was approximately the same in all groups and the groups were reasonably equated on a number of variables at the beginning of the study and after the loss of data (Table 2).

Predicting Benefit from the Lysergide Experience

Our assessment of the lysergide experience for each individual was compared to his drinking control at the end of 3 months and again at the end of a year to see if a relationship existed between any of the measures and his drinking control. Simple inspection of the data showed no relationship between any of our measures of a "good" or "poor" lysergide experience and benefit. Since these findings can be summarized very simply and unequivocally, the actual data are not presented.

Incidence of Lysergide Recrudescence

The 61 patients who gave information on the absence or presence of recrudescence were not prepared in the predischarge interview to provide this type of information. In response to our inquiry,

many described beneficial changes in attitudes, feelings and interpersonal relationships. These responses, along with those that definitely stated there were no effects, were tabulated as negative for recrudescence.

Our working definition of recrudescence was finally adopted after we read through all of the reports. We judged to be a valid recrudescence any perceptual thought or feeling experience which impressed the patient with its vividness and which was clearly related to the lysergide experience.

Of those who replied to this questionnaire, 19 patients (31%) reported at least one occasion of recrudescence during the year. There was a strong suggestion that drinking or withdrawal was a factor in triggering recrudescence. Patients reported this association and our tabulations showed that recrudescence was approximately twice as frequent in the drinking patients as in the abstainers. An additional group of 7 patients (11%) reported that they experienced alterations in dreaming. A single dream or several dreams were experienced as vivid, strongly colored, or a fragment of the lysergide experience relived. We regarded these dream reports as questionable examples of recrudescence. They are mentioned because altered dreams may be a little-recognized but real postlysergide effect.

If recrudescence occurs at all, the first occasion is usually within 3 or 4 months of the lysergide experience. Of the 19 patients who reported single or multiple recrudescences, 3 had the first within a few weeks of the experience, 12 (65%) reported the first on the 3-month follow-up questionnaire, 3 (15%) reported the first on the 6-month questionnaire, and 1 on the 12-month questionnaire.

There was a tendency for more than one recrudescence to occur. Four patients (22%) had only one experience, six (32%) reported two experiences, seven (36%) reported three experiences and two (10%) had four or more. Our lysergide-treated patients were composed of approximately 25% schizophrenic alcoholics. Simple inspection of the data indicated that there was no relationship between schizophrenia and the incidence of recrudescence.

Discussion

This investigation was essentially an effort to answer a practical treatment question about lysergide: Would the addition of a "single overwhelming lysergide experience" to the regular alcoholic rehabilitation program increase what was already being achieved?

We found among the lysergide-treated nonschizophrenic alcoholics a higher percentage who abstained from alcohol during each rating period, but the differences were not statistically significant and we could not conclude that lysergide was beneficial. Analysis of our data showed a statistically significant negative effect on the nonschizophrenic patient who volunteered for lysergide but did not receive it. We interpreted this as a "deprivation effect" that emphasizes the need for some type of control drug experience in future research. An alternative explanation for this apparent deprivation effect might be that our original reason for having a lysergide volunteer control group was valid and the volunteers are a self-selected group of patients who have a poor prognosis because of their strong propensity to value a drug cure for their problems. Staff members observed that the lysergide volunteers were slow "in getting involved" in the regular Program because they seemed to be waiting for the lysergide experience. If this was a significant influence in the study, the statistically significant difference between the nonschizophrenic lysergide-treated alcoholics and volunteer controls would reflect a real lysergide-related gain with less additive effect from the Program in both groups of volunteers.

Our data tend to corroborate those studies that did not screen out schizophrenic alcoholics and observed the negative effect on these patients both in the lysergide experience itself and later benefits (2, 4). The Blewett and Chwelos Scale 1 measured a statistically significant difference in the lysergide experience of schizophenic alcoholic patients.

Within the nonschizophrenic lysergide-treated group, the absence of a relationship between any of our measures of a "good" or "poor" experience and benefits might simply mean that our methods did not measure the crucial aspects of the experience that predict gain. The paradox might also be resolved by assuming that even a "bad" experience temporarily breaks up old patterns of feelings and behavior and makes the individual more plastic for new learning.

The major limitation of this study was created by the progressive loss of cases and the indeterminacy that this introduced. These losses do not necessarily invalidate the comparisons between groups but they tend to inflate the percentage of cases in the "total abstinence" and "much improved" categories in all the groups. The patients with whom contact was lost are probably not doing as well as the others and we can suspect that loss of contact occurred in

some cases because the patient regressed to uncontrolled drinking, became nomadic and abandoned his resolution to keep us informed. Our method of removing patients from the study who were rehospitalized or jailed for prolonged periods in the first 6 months of the follow-up also resulted in a population of patients at the end of the year from which some early failures had been culled. If we assume that none of the lost cases are abstaining, the percentages given in the "completely abstinent" category are reduced by 4 to 17%, depending on the particular group.

The study of recrudescence was conducted because of our opportunity to grossly quantify the incidence of these effects and to contribute something to the understanding of the minor and serious aftereffects of the lysergide treatment of alcoholics. Most of the literature on this subject has been reported as the "adverse effects," or "complications" or "untoward effects" of lysergide upon patients, research subjects and illicit users. An extensive review of these reports by Schwarz (14) shows that they consist almost entirely of examples of very serious complications including prolonged states of panic, confusion, psychosis, paranoia, and suicidal or homicidal behavior. None of our lysergide-treated patients, to the best of our knowledge, suffered any of these serious aftereffects. The cases we judged to be recrudescences were relatively brief experiences that varied from very uncomfortable to very pleasant, and occurred in about a third of the patients.

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